MILITARY SPECIFICATION

ALUMINUM ALLOY ARMOR ROLLED PLATE (1/4 TO 4 INCHES THICK), WELDABLE (ALLOY 2519)

1. SCOPE

1.1 Scope. This specification covers aluminum alloy 2519 armor rolled plate from 1/4 inch up to and including 4 inches in thickness.

1.2 Weldability. The material covered by this specification is weldable to itself and other weldable 2000 series alloys using a 2000 series filler and is weldable to 6000 series alloys using a 4000 series filler. It is not weldable to a 5000 series or 7000 series alloys by conventional welding techniques.

1.3 Classification. Material furnished under this specification shall be rolled plate in a T87 temper

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

AAA-PFE-1 - Aluminum Alloy Armor - Plate, Forged, Extruded
MIL-H-6088 - Heat Treatment of Aluminum Alloys

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-367 - Armor Test Data Reporting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials Technology Laboratory, ATTN: SLCMT-MSR-ES, Watertown, MA 02172-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC A3854

/FSC 9535/

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI - H35.2 - American National Standard Dimensional Tolerances for Aluminum Mill Products

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B557 - Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products

ASTM B666 - Identification Marking of Aluminum Products

ASTM D3951 - Practice for Commercial Packaging

ASTM E34 - Methods for Chemical Analysis of Aluminum and Aluminum Alloys

ASTM E35 - Practice for Sampling Wrought Nonferrous Metals and Alloys for Determination of Chemical Composition

ASTM E101 - Method for Spectrographic Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique

ASTM E227 - Method for Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique

ASTM E527 - Practice for Numbering Metals and Alloys (UNS)

ASTM E607 - Method for Optical Emmission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique, Nitrogen Atmosphere

ASTM E716 - Practice for Sampling Aluminum and its Alloys for Spectrochemical Analysis

ASTM G47 - Determining Susceptibility to Stress-Corrosion Cracking of High Strength Aluminum Alloy Products

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical associations specifications and standards are generallly available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 4.2.1, 6.2, and 6.6), a sample or samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.2.1. The approval of the first article sample(s) authorizes the commencement of shipment but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples and test plates shall be manufactured by the process proposed for use on production armor. When specified in the contract or order a certified test report of the declared chemical analysis shall be be prepared (see 6.2.2) and submitted to the contracting agency and to the ballistic test agency. The ballistic test agency will record the first article ballistic test plates submitted, showing the dates tested. Requests from the procuring activity to the ballistic test agency as to prior conformance with first article tests must be accompanied by copies of the first article test firing record. Any deviation(s) noticed by the ballistic agency will be brought to the attention of the contracting agency and to the manufacturer.

3.1.1 First time producer. First time producers wishing to qualify for this specification should follow the instructions of 6.5.

3.1.2 Process changes. After an armor material has successfully complied with the requirements of this specification, any deliberate change in the mill processing by the manufacturer which could affect the material characteristics controlled by this specification, shall be drawn to the attention of the procuring activity and the contractor. Tests, as stated in this specification, may be required to determine if the material made with process changes meets the requirements of this specification (see 6.2.2).

3.2 Chemical composition. Chemical composition of material furnished under this specification shall be within the limits specified in table I (see 6.2.2).

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of rolled plate shall meet the requirements of table II, when tested as specified in 4.6.2.

3.3.1 If mechanical property requirements differ from the values shown in table II, or if any other properties are required, the ballistic requirements shall be as agreed upon between the procuring activity and the manufacturer.

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3.4 Ballistic requirements. Each ballistic test plate shall be proof fired for resistance to penetration by determining a protection ballistic limit, BL(P) (see 6.2.2). The BL(P) shall be as specified in appendix A. No ballistic limit shall be specified for plate thickness of 0.250 to 0.500 inch inclusive.

TABLE I. Chemical composition, percent (ladle analysis).

Element	Min.	Max.
Silicon	-	0.25 1/
Iron	-	0.30 1/
Copper	5.30	6.40
Manganese	0.10	0.50
Magnesium	0.05	0.30
Titanium	0.02	0.10
Zinc	0.02	0.10
Zirconium	0.10	0.25
Vanadium	0.05	0.15
Others, each	-	0.05
Others, total	-	0.15
Aluminum, by difference	Remainder	

1/ The combined silicon and iron content shall not exceed 0.40 percent.

TABLE II. Mechanical properties of plate. 1/

Direction	Thickness, inches	Tensile strength min ksi.	Yield strength 0.2% offset min ksi	Elongation in 2 inches min. percent
Long transverse	0.250-1.000	66.0	58.0	6
	1.001-2.000	66.0	58.0	5
	2.001-3.000	66.0	58.0	4
	3.001-4.000, incl.	66.0	58.0	3

1/ Limits are tentative minima per Aluminum Association Registration 27 March 1986.

3.5 Stress-corrosion resistance. When stress corrosion testing is specified in the contract or order (see 6.2), plate 0.75 inch and greater in thickness shall be resistant to stress corrosion cracking, with 3 out of 3 specimens showing no evidence of cracking when tested in accordance with 4.6.3.

3.6 Thermal processing. Heat treatment shall conform to the requirements of MIL-H-6088 and shall be such as to enable the material to meet the requirements of this specification.

3.7 Dimensions and tolerances.

3.7.1 Dimensions. Size of plate shall be as specified in the invitation for bids, contract or order (see 6.2).

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3.7.2 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates submitted for acceptance shall not vary from the specified dimensions by an amount greater than the tolerances shown in ANSI H35.2. Similarly, the thickness tolerance shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered thickness, inches	Tolerance, inch Minus	Plus
0.250-0.499, incl.	0.020	--
0.500-0.875, incl.	0.030	--
0.876-1.125, incl.	0.035	--
1.126-1.375, incl.	0.040	--
1.376-1.625, incl.	0.045	--
1.626-1/875, incl.	0.050	--
1.876-2.250, incl.	0.055	--
2.251-2.750, incl.	0.060	--
2.751-3.000, incl.	0.065	--
3.001-4.000, incl.	0.070	--

1/ The values for the plus column are derived from the full range tolerances specified in table 3.1 of ANSI 35.2. For example, the a 2.0 inch thick by a 73 inches wide plate would be as follows:

Thickness tolerance (from ANSI H 35.2)	±0.0623
Full tolerance less minus tolerance	0.125 - 0.055 = 0.070
Derived plus tolerance	0.070
Tabulated minus tolerance	0.055

3.8 Identification marking. Unless otherwise specified (see 6.2) each plate shall be continuously line marked along the rolling direction in accordance with ASTM B666.

3.8.1 Ballistic test plates. In addition to the marking in 3.8, each ballistic test plate shall be marked with the letters PRE for first article test plates and ACC for acceptance test plates. This marking shall be impression stamped or stenciled on the plate rolled surface not less than 1/2 inch high and within one inch of the plate edge.

3.9 Workmanship. Rolled plate shall be uniform in quality and shall be clean, smooth and commercially free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use in fabrication. Surface cracks, edge cracks or edge laminations shall be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Ballistic tests shall be performed by Aberdeen Proving Ground (see 6.2).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.2.1 First article inspection. First article inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.6.1).
(b) Mechanical properties (see 3.3 and 4.6.2).
(c) Ballistic tests (see 3.4, 4.6.4 and Appendix A).
(d) Stress corrosion tests (see 3.5 and 4.6.3.)
(e) Dimensions and tolerances (see 3.7 and 4.7.1).
(f) Identification marking (see 3.8 and 4.7.2).
(g) Workmanship (see 3.10 and 4.7.2).

4.2.2 Quality conformance inspection. Lot acceptance inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.6.1).
(b) Mechanical properties (see 3.3 and 4.6.2).
(c) Ballistic tests (see 3.4, 4.6.4 and Appendix A).
(d) Dimensions and tolerances (see 3.7 and 4.7.1).
(e) Identification marking (see 3.8 and 4.7.2).
(f) Workmanship (see 3.10 and 4.7.2).
(g) Preparation for shipment (see section 5 and 4.7.3).

4.3 Lot. For rolled plate a lot shall consist of plate of the same ordered thickness and temper submitted for inspection as a unit, provided that the weight of the finished plate in the lot does not exceed 50,000 pounds.

4.4 Sampling.

4.4.1 Sampling for first article testing.

4.4.1.1 For chemical composition. Samples for chemical analysis shall be removed from each plate selected for ballistic testing.

4.4.1.2 For mechanical properties tests: A tension test specimen for first article tests shall be removed from the same plate material that has been selected for the ballistic test. The location and type specimen shall be as specified in 4.5.2.

4.4.1.3 Stress corrosion tests. Stress corrosion test specimens shall be removed from the same plate selected for the ballistic test. The type of specimens shall be as specified in 4.6.3.

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4.4.1.4 Ballistic tests: Two plates, 12 by 36 inches by the ordered thickness, shall be submitted for ballistic testing. The orientation of these plates with respect to the rolling direction shall be at the option of the manufacturer. The ballistic test may be waived at the discretion of the procuring activity if the manufacturer within 37 months has produced acceptable plate within plus and minus 0.250 inch of the ordered thickness, provided also that the test conditions are the same as for previously accepted plate.

4.4.2 Sampling for quality conformance testing.

4.4.2.1 Chemical composition:

(a) For ladle analysis. At least one sample shall have been taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the Government at the ingot producer's facility.

(b) For check analysis. When sampling has not been made in accordance with 4.4.2.1(a), one sample shall be taken for each 4,000 pounds or less in each lot in accordance with methods specified in 4.6.1.

4.4.2.2 Mechanical properties tests: From each lot samples for tension tests shall be selected in accordance with table IV. Each sample shall be selected from a different plate in the lot. Should a lot consist of only one plate, only one tension test sample shall be taken. One tension test specimen shall be made from each sample.

TABLE IV. Number of tension test samples.

Lot size, pounds	Minimum number of samples
To 8,000, incl.	2
8,001-12,000, incl.	3
12,001-20,000, incl	4
Over 20,000, incl.	5

4.4.2.3 Stress corrosion tests (for surveillance testing only). One plate, approximately 12 inches by the ordered thickness shall be selected from the lot for stress corrosion tests. At the discretion of the procuring activity, reduced sampling may be instituted, i.e., not every lot need be sampled (see 4.9).

4.4.2.4 Ballistic testing: One plate approximately 12 inches by 36 inches by the ordered thickness shall be selected from the lot for ballistic testing. The orientation of this plate with respect to the rolling direction shall be at the option of the manufacturer.

4.5 Test specimen preparation.

4.5.1 Chemical analysis. Specimens for chemical analysis shall be prepared in accordance with the applicalble method specified in 4.6.1.

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4.5.2 Mechanical properties: Tension test specimens shall conform to the dimensions of the standard specimens (figure 8) of ASTM B557. They shall be taken in the long transverse direction. From plate in nominal thickness 0.25 to 1.50 inches inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces; from plate in nominal thicknesses greater than 1.50 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.5.3 Stress corrosion. The stress corrosion specimens shall be stressed at 75% of the specified yield strength. The location of the specimens taken from the plate shall be specified in the drawing (see 6.2).

4.6 Test procedures.

4.6.1 Chemical composition. The determination of the chemical composition shall be made in accordance with one or more of the following test methods: ASTM E34, E101, E227, E607.

4.6.2 Mechanical properties. The mechanical properties shall be determined in accordance with ASTM B557.

4.6.3 Resistance to stress corrosion. When stress corrosion testing is specified, the test shall be conducted in accordance with ASTM G47.

4.6.4 Ballistic tests. Plates shall be tested using projectiles chosen in accordance with the thickness ranges shown in table V. Plate thickness as determined by the ballistic test agency shall be used to determine the required ballistic limit for that plate. Individual thickness measurements are to be read from a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 inch and the average of these readings reported to the nearest 0.005 inch. Measurements shall be made on the intended impact area.

TABLE V. Acceptance ballistic test plates.

Ordered or nominal design thickness specified on drawings, inches	Test angle	Projectile to be used
0.500- .749	30°	Cal. .30 AP, M2
0.750-0.950	0°	Cal. .50 FS
0.951-1.550	0°	20 mm FS
0.951-1.550	0°	Cal. .30 AP, M2
1.551-3.000	0°	Cal. .50 AP, M2
(a) 3.001-4.000	0	14.5 mm API, BS41

(a) Ballistic limit requirements to be developed.

4.7 Examination.

4.7.1 Dimensions. Each plate within the lot shall be measured to determine compliance with the requirements for dimensions and tolerances (see 3.7) or the lot shall be sampled in accordance with sampling procedures approved by the procuring activity.

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4.7.2 Visual. Each plate shall be examined to determine compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.7.3 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

4.8 Rejection and retest.

4.8.1 First article. Failure of the first article test plates to meet the requirements of 4.2.1 indicates rejection of the product and process.

4.8.1.1 Retests. Resubmission of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfation of the procuring activity. Samples submitted for retest shall be marked R1, R2 and R3, if applicable, in addition to the requirements of 3.8.

4.8.2 Acceptance.

4.8.2.1 Examination. A quality conformance lot shall be rejected for failure to meet the visual, dimensional and preparation for delivery requirements when examined in accordance with 4.7.

4.8.2.2 Tests. A quality conformance lot shall be rejected for failure to meet any of the requirements of any of the tests outlined in 4.2.2.

4.8.2.3 Retests.

4.8.2.3.1 Chemical composition and mechanical properties. Retests shall be conducted in accordance with the methods specified in 4.6.1 and 4.6.2.

4.8.2.3.2 Stress corrosion (first article) testing only. If the manufacturer so desires, a retest shall be performed using three additional specimens. No cracking shall occur in 10 days. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must conform to the requirements outlined in 4.6.3.

4.8.2.3.3 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A-50.2.

4.8.2.4 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product, meeting the testing requirements is being produced and provided the manufacturer agrees to maintain the same manufacturing procedures.

5. PACKAGING

5.1 Preservation.

5.1.1 Commercial. The commercial preservation of material shall be in accordance with the requirements of D3951.

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5.2 Packing.

5.2.1 Commercial. The preserved material shall be packed in accordance with the requirements of D3951.

5.3 Marking of shipment. In addition to any special marking specified in the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material specified herein is for military vehicles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) If first article inspection is required (see 3.1).
(c) If stress corrosion is required (see 3.5).
(d) Drawings, dimensions and tolerances (see 3.7, table III and 4.6.3.
(e) If different identification marking is required (see 3.8).
(f) Destination of ballistic test samples (see 4.1.1).
(g) Additional marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
3.1.3	Report, First Article Test	UDI-T-23790	
3.2, 3.3	Certification Data Report	UDI-T-23264	
3.4	Armor Material Test Reports	DI-MISC-80073	Format II

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Manufacturer. The manufacturer is defined as the company producing the aluminum alloy plate.

6.3.2 Contractor. The contractor is the company which has a direct contract from the Government to furnish an end item - usually a vehicle. Also known as the prime contractor.

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6.3.3 Contracting officer. The term "contracting officer" means the person executing a contract on behalf of the Government and any other officer or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of a contracting officer acting within the limits of his authority.

6.3.4 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.4 Mechanical properties and ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure aluminum armor material meeting the specified ballistic requirements (see 3.4).

6.5 Potential suppliers. Potential suppliers who have not previously supplied armor plate to MIL-A-46192(MR) and wish to have their material ballistically tested, may do so at their own expense. It is recommended that inquiries for such testing be directed to Commander US Army CSTA, ATTN: AMSTE-TO-O, Aberdeen Proving Ground, MD 21005-5005.

6.6 New contracts sponsored by government agencies. At the time that a new contract is initiated for the production of combat vehicles, the contractor's supplier is to estimate for the contractor the number, size and delivery schedule of the ballistic test plates which are to be submitted for first article acceptance testing. A lead time of 60 days after the contract has been signed is to be allowed prior to shipment of the first ballistic test plate(s) to APG to ensure that all administrative functions for the establishment of a new CSTA project have been completed in preparation for the test. The contracting government agency is to initiate the new project through a letter to Commander, US Army CSTA, ATTN: AMSTE-TO-O, APG, MD 21005-5005 requesting a cost estimate for the ballistic testing of the applicable number and sizes of plates. In the case of increases in scope of existing projects, similar correspondence is needed.

6.7 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

To go from English	Multiply by	To get Metric SI Unit
inch	0.0254	metre (m)
foot	0.3048	metre (m)
foot/lb	1.3558	joule (J)
feet/sec	0.3048	metre per second (m/s)
pounds/sq. inch	0.00689	mega pascal (mpa)

Note: Conversion factors can be obtained from ASTM E 380 entitled "Metric Practice Guide".

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Custodian:

Army - MR

Review Activities:

Army - AT, AR, ME, TE
DLA - IS

Preparing activity:

Army - MR

Project No. 9535-A029

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APPENDIX A

BALLISTIC TESTING OF ALUMINUM ALLOY ARMOR PLATE

A-10 SCOPE. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor, plate, weldable, when tested in accordance with the provisions of this specification. This Appendix is a mandatory part of this specification.

A-20 APPLICABLE DOCUMENTS

A-20.1 The following document forms a part of this appendix:

Proving Ground Acceptance Test Procedure No. AAA-PFE-1 25 June 1979, for Aluminum Alloy Armor-Plate, Forged, Extruded.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

A-30 DEFINITIONS

A-30.1 Fair impact. A fair impact results when an unyawed projectile strikes an unsupported area of the ballistic test plate sample at such a location that there is at least two calibers of undisturbed metal between this impact and any previous impact, hole, crack, edge of plate or spalled area.

A-30.2 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052 H36 aluminum alloy (or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches behind and parallel to the test plates or other ballistic sample.

A-30.3 Protection complete penetration, CP(P). A protection complete penetration will have been obtained when the projectile or a fragment of the projectile or a fragment of the ballistic test plate is thrown beyond the rear face of the plate with sufficient energy to perforate an approximate 0.020 inch thick sheet of 2024-T3 aluminum alloy (or a 0.014 inch thick sheet of 5052 H36) placed parallel to and 6 inches behind the plate.

A-30.4 Partial penetration, PP(P). Any fair impact that is not a complete penetration shall be considered a partial penetration.

A-30.5 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

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A-40 REQUIREMENTS

A-40.1 Resistance to penetration. The minimum required V_{50} ballistic limit shall be in accordance with the values shown in tables A-I through A-IV.

A-50 TESTS

A-50.1 Ballistic tests. Testing shall be in accordance with Proving Ground Acceptance Test Procedure AAA-PFE-1, 25 June 1979, for Aluminum Alloy Armor-Plate, Forged, Extruded, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A-50.1.1 Temperature conditioning. Prior to the test, the test item(s) will be temperature conditioned at least eight hours. Thermostatic control will be such that the average temperature of the item during the test shall be $72^{\circ} \pm 15^{\circ}F$ ($22^{\circ} \pm 8^{\circ}C$).

A-50.1.2 Protection ballistic limit, BL(P).

A-50.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) will be reported.

A-50.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit will be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing will continue until a 10-round ballistic limit has been attained using the smallest practical velocity spread. Ten round ballistic limits will be reported as such on the armor data forms.

A-50.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit which has been attained is within $\pm$ 10 fps of the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, will be fired to reduce the gap to 25 fps or less. The ballistic limit will then be recomputed using the above criteria. The recomputed BL(P) will be reported as the BL(P) of the plate. In borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P).

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A-50.2 Rejection and retest of ballistic plates.

A-50.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A-50.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material as acknowledge by the procuring activity. Two retest plates shall be submitted for first article testing and both tests must pass; otherwise, the armor material shall be rejected.

A-50.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates rejection of the lot; however, the final decision will depend on the outcome of retests, if submitted.

A-50.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure, may submit at his own expense two additional test plates from the same lot for ballistic retest. If either of these plates fail the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fail, the lot shall be permanently rejected.

A-50.3 Disposition of ballistic test plates.

A-50.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates will be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A-50.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates will be returned, upon request, as in A-50.3.1.

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APPENDIX A

TABLE A-I. Minimum Required Ballistic Limits - Caliber .30 AP M2 Projectile at 30° Obliquity.

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.470	1398	0.595	1698	0.720	1955
0.475	1411	0.600	1709	0.725	1964
0.480	1424	0.605	1720	0.730	1974
0.485	1437	0.610	1731	0.735	1983
0.490	1450	0.615	1742	0.740	1993
0.495	1462	0.620	1752	0.745	2002
*0.500	1475	0.625	1763	**0.749	2009
0.505	1488	0.630	1774	0.750	2011
0.510	1500	0.635	1784	0.755	2021
0.515	1512	0.640	1795	0.760	2030
0.520	1525	0.645	1805	0.765	2039
0.525	1537	0.650	1815	0.770	2048
0.530	1549	0.655	1826	0.775	2057
0.535	1561	0.660	1836	0.780	2067
0.540	1573	0.665	1846	0.785	2075
0.545	1585	0.670	1856	0.790	2085
0.550	1596	0.675	1866	0.795	2094
0.555	1608	0.680	1876	0.800	2103
0.560	1619	0.685	1886	0.805	2112
0.565	1631	0.690	1896	0.810	2120
0.570	1642	0.695	1906	0.815	2129
0.575	1654	0.700	1916	0.820	2138
0.580	1665	0.705	1926	0.825	2147
0.585	1676	0.710	1935	0.828	2152
0.590	1687	0.715	1945	0.830	2156

* Specification requirements begin for ordered thickness .500 inch.

** Specification requirements end for ordered thickness .749 inch.

Numbers above and below upper and lower limits are requirements for plates within the tolerance limits.

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APPENDIX A

TABLE A-II Minimum required ballistic limits - .30 Caliber AP M2 Projectile at 0 Degrees Obliquity

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.720	1778	0.920	2069	1.120	2324
0.725	1786	0.925	2076	1.125	2330
0.730	1794	0.930	2082	1.130	2336
0.735	1801	0.935	2089	1.135	2342
0.740	1809	0.940	2096	1.140	2348
0.745	1817	0.945	2103	1.145	2354
*0.750	1824	0.950	2109	1.150	2360
0.755	1832	0.955	2116	1.155	2365
0.760	1840	0.960	2122	1.160	2372
0.765	1847	0.965	2129	1.165	2378
0.770	1855	0.970	2136	1.170	2384
0.775	1862	0.975	2142	1.175	2390
0.780	1870	0.980	2149	1.180	2396
0.785	1877	0.985	2155	1.185	2402
0.790	1885	0.990	2162	1.190	2407
0.795	1892	0.995	2168	1.195	2413
0.800	1900	1.000	2175	1.200	2419
0.805	1907	1.005	2181	1.205	2425
0.810	1914	1.010	2187	1.210	2431
0.815	1922	1.015	2194	1.215	2436
0.820	1929	1.020	2200	1.220	2442
0.825	1936	1.025	2207	1.225	2448
0.830	1943	1.030	2213	1.230	2454
0.835	1950	1.035	2219	1.235	2459
0.840	1958	1.040	2226	1.240	2465
0.845	1965	1.045	2232	1.245	2471
0.850	1972	1.050	2238	1.250	2476
0.855	1979	1.055	2244	1.255	2482
0.860	1986	1.060	2251	1.260	2488
0.865	1993	1.065	2257	1.265	2493
0.870	2000	1.070	2263	1.270	2499
0.875	2007	1.075	2269	1.275	2505
0.880	2014	1.080	2275	1.280	2510
0.885	2021	1.085	2282	1.285	2516
0.890	2028	1.090	2288	1.290	2521
0.895	2035	1.095	2294	1.295	2527
0.900	2042	1.100	2300	1.300	2532
0.905	2049	1.105	2306	1.305	2538
0.910	2055	1.110	2312	1.310	2544
0.915	2062	1.115	2318	1.315	2549

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APPENDIX A

TABLE A-II. Minimum required ballistic limits -.30 caliber AP M2 projectile at 0 degrees obliquity)(Continued).

Thickness (inch)	Required BL(P) fps		Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
1.320	2555		1.425	2668	1.530	2775
1.325	2560		1.430	2673	1.535	2781
1.330	2566		1.435	2678	1.540	2785
1.335	2571		1.440	2683	1.545	2791
1.340	2576		1.445	2689	1.550	2796
1.345	2582		1.450	2694	1.555	2301
1.350	2587		1.455	2699	1.560	2306
1.355	2593		1.460	2704	1.565	2311
1.360	2598		1.465	2709	1.570	2315
1.365	2604		1.470	2715	1.575	2321
1.370	2609		1.475	2720	1.580	2326
1.375	2614		1.480	2725	1.585	2331
1.380	2620		1.485	2730	1.590	2336
1.385	2625		1.490	2735	1.595	2341
1.390	2630		1.495	2740	1.600	2346
1.395	2636	**	1.500	2746	1.605	2851
1.400	2641		1.505	2751	1.610	2856
1.405	2646		1.510	2756	1.615	2861
1.410	2652		1.515	2761	1.620	2866
1.415	2657		1.520	2766	1.625	2871
1.420	2662		1.525	2771		

* Specification requirements begin for ordered thickness .750 inch.
**Specification requirements end for ordered thickness 1.500 inch.

Numbers above and below upper and lower limits are requirements for plates within the tolerance limits.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits .50 caliber fragment simulating projectile at 0 degrees obliquity.

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.720	1827	0.825	2274	0.930	2823
0.725	1847	0.830	2297	0.935	2852
0.730	1866	0.835	2321	0.940	2881
0.735	1886	0.840	2345	0.945	2911
0.740	1905	0.845	2370 **	0.950	2941
0.745	1925	0.850	2394	0.955	2971
* 0.750	1946	0.855	2419	0.960	3002
0.755	1966	0.860	2444	0.965	3033
0.760	1987	0.865	2469	0.970	3064
0.765	2007	0.870	2495	0.975	3095
0.770	2028	0.875	2521	0.980	3127
0.775	2050	0.880	2547	0.985	3159
0.780	2071	0.885	2573	0.990	3192
0.785	2093	0.890	2600	0.995	3225
0.790	2114	0.895	2627	1.000	3258
0.795	2136	0.900	2654	1.005	3291
0.800	2159	0.905	2681	1.010	3325
0.805	2181	0.910	2709	1.015	3359
0.810	2204	0.915	2737	1.020	3394
0.815	2227	0.920	2765	1.023	3414
0.820	2250	0.925	2794	1.025	3423

* Specification requirements begin for this ordered thickness .750 inch.
** Specification requirements end for ordered thickness .950 inch.

Numbers above and below upper and lower limits are requirements for plates within the tolerance limits.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits 20 mm fragment simulating projectile at 0 degrees obliquity.

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
0.915	1283	1.110	1677	1.310	2198
0.920	1292	1.115	1689	1.315	2213
0.925	1301	1.120	1700	1.320	2228
0.930	1310	1.125	1712	1.325	2243
0.935	1319	1.130	1724	1.330	2255
0.940	1329	1.135	1735	1.335	2273
0.945	1338	1.140	1747	1.340	2388
0.950	1347	1.145	1759	1.345	2304
*0.951	1349	1.150	1771	1.350	2314
0.955	1356	1.155	1783	1.355	2335
0.960	1366	1.160	1795	1.360	2350
0.965	1375	1.165	1808	1.365	2365
0.970	1385	1.170	1820	1.370	2382
0.975	1394	1.175	1832	1.375	2492
0.980	1404	1.180	1845	1.380	2414
0.985	1414	1.185	1857	1.385	2430
0.990	1423	1.190	1870	1.390	2447
0.995	1433	1.195	1883	1.395	2463
1.000	1443	1.200	1895	1.400	2479
1.005	1453	1.205	1908	1.405	2496
1.010	1463	1.210	1921	1.410	2513
1.015	1473	1.215	1934	1.415	2530
1.020	1483	1.220	1947	1.420	2547
1.025	1493	1.225	1960	1.425	2564
1.030	1504	1.230	1974	1.430	2581
1.035	1514	1.235	1987	1.435	2598
1.040	1524	1.240	2000	1.440	2615
1.045	1535	1.245	2014	1.445	2633
1.050	1546	1.250	2028	1.450	2650
1.055	1556	1.255	2041	1.455	2663
1.060	1567	1.260	2055	1.460	2686
1.065	1578	1.265	2069	1.465	2704
1.070	1588	1.270	2083	1.470	2722
1.075	1599	1.275	2097	1.475	2740
1.080	1610	1.280	2111	1.480	2758
1.085	1621	1.285	2129	1.485	2777
1.090	1632	1.290	2120	1.490	2795
1.095	1643	1.295	2154	1.495	2614
1.100	1655	1.300	2169	** 1.500	2633
1.105	1666	1.305	2183	1.505	2852

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits 20 mm fragment simulating projectile at 0 degrees obliquity (Continued).

Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps	Thickness (inch)	Required BL(P) fps
1.510	2871	1.550	3027	1.590	3192
1.515	2890	1.555	3047	1.595	3213
1.520	2909	1.560	3067	1.600	3234
1.525	2928	1.565	3088	1.605	3256
1.530	2948	1.570	3108	1.610	3277
1.535	2967	1.575	3129	1.615	3299
1.540	2987	1.580	3150	1.620	3321
1.545	3007	1.585	3171	1.625	3343

* Specification requirements begin for ordered thickness 0.951 inch.
** Specifiction requiremtns end for ordered thickness 1.500 inches.

Numbers above and below upper and lower limits are requirements for plates within the tolerance limits.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-46192(MR)	ALUMINUM ALLOY ARMOR ROLLED PLATE(1/4 TO 4 INCHES THICK), WELDABLE (ALLOY 2519)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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